

MCS-218

DATA COMMUNICATION AND COMPUTER NETWORKS

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Qus 1.

**(a) What is meant by CRC ? Write the following bitstring in polynomial representation :
4 "1100010"**

Ans :

CRC stands for Cyclic Redundancy Check.

✓ It is an **error-detection technique** used in data communication and storage to detect accidental changes (errors) in raw data.

✓ In CRC, the data bits are treated as coefficients of a **binary polynomial**, which is divided (modulo-2 division) by a **generator polynomial**.

✓ The **remainder** of this division is the **CRC code**, which is appended to the message for transmission.

✓ At the receiver's end, the same division is performed to check if the remainder is **zero**.
If it is zero → **No error detected**.
Otherwise → **Error detected**.

Convert the bitstring 1100010 to polynomial representation

Let's write the bitstring clearly:

Bitstring: 1 1 0 0 0 1 0

Each bit corresponds to a coefficient of a polynomial in descending powers of (x):

- The **leftmost bit** corresponds to (x^6) (since there are 7 bits total $\rightarrow$ powers go from 6 down to 0).
- A **1** means the term is present, a **0** means it is absent.

So:

$$[\quad 1100010 \rightarrow 1 \cdot x^6 + 1 \cdot x^5 + 0 \cdot x^4 + 0 \cdot x^3 + 0 \cdot x^2 + 1 \cdot x^1 + 0 \cdot x^0]$$

Now remove the zero terms:

$$[\quad P(x) = x^6 + x^5 + x]$$

✓ **Final Answer:**

- **Meaning of CRC:** Cyclic Redundancy Check — an error detection method based on polynomial division.
- **Polynomial form of 1100010:**

$$[\quad P(x) = x^6 + x^5 + x]$$

(b) What are Wireless LANs ? Discuss the disadvantages of using radio transmitters.

Ans :

① **What are Wireless LANs (WLANs)?**

A **Wireless Local Area Network (WLAN)** is a type of **local area network** that uses **radio waves** (wireless communication) instead of cables to connect devices such as computers, smartphones, printers, and other network equipment.

✓ Key points:

- A WLAN provides the **same network services** as a wired LAN — such as file sharing, internet access, and resource sharing — but **without physical cables**.
- It typically uses **Wi-Fi (IEEE 802.11 standards)** for communication.
- Devices connect through a **Wireless Access Point (WAP)**, which acts as a hub connecting wireless devices to the wired network (or directly to each other in ad-hoc mode).

② Disadvantages of using radio transmitters (in WLANs)

Although WLANs are convenient, using **radio transmitters** for communication introduces several drawbacks:

(i) Security Issues

- Radio signals can travel beyond the physical boundaries of a building.
- Unauthorized users can intercept signals, leading to **data theft, eavesdropping, or unauthorized access**.
- Extra security measures (encryption, firewalls) are needed.

(ii) Interference

- Radio transmitters are affected by **interference** from other electronic devices (like microwaves, cordless phones, Bluetooth devices).
- This can cause **signal degradation, loss of data, or network slowdowns**.

(iii) Limited Range and Coverage

- WLAN signals have a **limited range** — typically 30–100 meters indoors.
- Obstacles such as **walls, metal objects, and furniture** weaken signals.
- Requires multiple access points for large areas.

(iv) Lower Data Transfer Speed

- Wireless connections usually offer **lower bandwidth** and **higher latency** compared to wired LANs.
- This affects performance, especially for large data transfers or real-time applications.

(v) Health and Safety Concerns

- Continuous exposure to radio frequency (RF) radiation, though low, raises **health concerns** among some users.

✓ Summary Answer

Aspect	Explanation
Wireless LAN (WLAN)	A local area network that uses radio signals instead of cables for data communication.
Disadvantages of Radio Transmitters	1. Security risks (signal interception) 2. Signal interference 3. Limited range 4. Lower speed 5. Possible health concerns

Would you like me to also include **advantages of WLANs** (for a balanced exam-style answer)?

(c) What is Transmission Media ? Compare optical fiber with copper wire.

Ans :

① What is Transmission Media?

Transmission Media refers to the **physical pathway** or **channel** through which data is transmitted from one device to another in a computer network.

It is the **medium** that carries **electromagnetic signals** representing data between a transmitter and a receiver.

✓ Types of Transmission Media:

1. Guided Media (Wired Media):

Data signals are guided along a solid medium.

Examples: **Twisted pair cable, Coaxial cable, Optical fiber.**

2. Unguided Media (Wireless Media):

Data is transmitted through air or vacuum using electromagnetic waves.

Examples: **Radio waves, Microwaves, Infrared.**

② Comparison: Optical Fiber vs Copper Wire

Feature	Optical Fiber	Copper Wire (Twisted Pair / Coaxial)
Transmission Medium	Uses light signals for data transmission.	Uses electrical signals for data transmission.
Material	Made of glass or plastic fibers.	Made of copper metal .
Bandwidth / Data Rate	Very high bandwidth — supports Gbps to Tbps speeds.	Limited bandwidth — supports up to a few hundred Mbps.
Signal Loss (Attenuation)	Low attenuation — signals can travel long distances without amplification.	High attenuation — requires repeaters for long distances.
Interference	Immune to electromagnetic interference (EMI) because it uses light.	Susceptible to EMI and radio frequency interference.
Security	Difficult to tap — very secure against eavesdropping.	Easier to tap or intercept signals.
Cost	Higher installation cost , but lower maintenance.	Cheaper to install, but more maintenance required.
Durability	Fragile (glass) and needs careful handling.	More robust and flexible.

Weight

Lightweight and thin.

Heavier and bulkier.

✓ Summary

- **Transmission Media** = The path through which data travels from sender to receiver.
- **Optical Fiber** is superior in speed, distance, and security, while **Copper Wire** is cheaper and easier to install but slower and more prone to interference.

(d) What is meant by burst error ? How can burst errors be corrected ?

Ans :

A **burst error** is a type of **data transmission error** in which **two or more bits in a sequence** of bits are **altered (flipped)** during transmission.

In other words,

A burst error occurs when **a group of consecutive bits** in the transmitted data gets corrupted.

✓ Example:

Original data:

1100110011

Received data:

1111110011

Here, bits 2 to 5 are changed → this is a **burst error of length 4**.

2 Burst Error Characteristics

- The error affects a **series of bits**, not just one bit.
- Burst length = the number of bits **from the first corrupted bit to the last corrupted bit** (including them).
- Common in **noisy channels**, especially wireless or magnetic storage media.

3 How can Burst Errors be Corrected?

There are several methods to **detect and correct burst errors** depending on the system's design.

(i) Cyclic Redundancy Check (CRC)

- CRC can **detect** burst errors very effectively.
- A properly chosen generator polynomial can detect:
 - All single-bit errors
 - All double-bit errors
 - All odd-numbered errors
 - **All burst errors up to the length of the CRC bits**

However, **CRC only detects** errors — it doesn't correct them.

(ii) Interleaving

- The data is arranged in a **matrix form** and transmitted **row by row**, but received **column by column** (or vice versa).
- This spreads the burst error over **multiple codewords**, turning one long burst into **several single-bit errors**.
- Then, **single-bit error correcting codes** (like Hamming codes) can fix them.

✓ Example:

If a burst error of 8 bits occurs, interleaving may spread it as 1-bit errors in 8 different blocks — which can then be corrected easily

(iii) Error Correction Codes (ECC)

- **Block codes** (like **Reed–Solomon codes**) and **Convolutional codes** are used to **detect and correct** burst errors.

- **Reed–Solomon codes**, used in CDs, DVDs, and QR codes, can correct long bursts of errors effectively.

✓ Summary Table

Aspect	Explanation
Burst Error	A group of consecutive bits in a data unit are corrupted during transmission.
Cause	Noise, interference, or faulty communication lines.
Detection / Correction Methods	1. CRC (detects burst errors) 2. Interleaving (spreads burst into single-bit errors) 3. Reed–Solomon / ECC (corrects burst errors)

✓ In short:

Burst Error: Two or more bits in a sequence get corrupted during transmission.

Correction: Use **interleaving** and **error correction codes (like Reed–Solomon)** to correct them.

(e) Explain the three types of internetwork addresses with a suitable example for each.

Ans :

① What is an Internetwork Address?

An **internetwork address** (or **IP address**) is a **logical address** assigned to every device connected to a network that uses the **Internet Protocol (IP)** for communication.

It uniquely identifies a **host** or **network interface** and helps data packets find their **source and destination** across interconnected networks.

Internetwork addresses are divided into **three types** based on how they identify destinations:

② Types of Internetwork Addresses

(i) Unicast Address

Meaning:

A **Unicast address** identifies a **single, specific host** on a network.

➡ Data sent to a unicast address is delivered **only to that one destination**.

Example:

If Computer A (192.168.1.10) sends a message to Computer B (192.168.1.20), the destination IP **192.168.1.20** is a **unicast address**.

✅ **Example in IPv4:** 192.168.1.5

✅ **Example in IPv6:** 2001:db8::1

Used for:

- Normal one-to-one communication between devices.
- Web browsing, email, file transfers, etc.

(ii) Broadcast Address

Meaning:

A **Broadcast address** is used to send data to **all hosts** on a **local network segment**.

➡ Data sent to a broadcast address is received by **every device** in that network.

Example:

For a network with address 192.168.1.0 and subnet mask 255.255.255.0, the broadcast address is **192.168.1.255**.

✅ **Example:**

If a message is sent to 192.168.1.255, all devices in the 192.168.1.x network receive it.

Used for:

- Network discovery (e.g., ARP requests)
- DHCP (Dynamic Host Configuration Protocol) broadcast messages

(iii) Multicast Address

Meaning:

A **Multicast address** identifies a **group of hosts** that are interested in receiving the same data.

➡ Data sent to a multicast address is delivered **only to group members**, not to everyone.

Example:

IPv4 multicast addresses range from **224.0.0.0 to 239.255.255.255**.

Example: **224.0.0.1** – all systems on the local network segment.

✓ **Example in IPv4:** **224.0.0.5** (used by OSPF routers)

✓ **Example in IPv6:** **ff02::1** (all nodes on the local link)

Used for:

- Video conferencing, IPTV, streaming, group communication protocols.

③ Summary Table

Type	Purpose	Delivery	Example	Used for
Unicast	One-to-one communication	To a single host	192.168.1.10	Web, email, FTP
Broadcast	One-to-all communication	To all hosts in a network	192.168.1.255	ARP, DHCP
Multicast	One-to-many communication	To a specific group	224.0.0.1	Video streaming, conferencing

✓ In short:

- **Unicast:** One-to-one
- **Broadcast:** One-to-all

- **Multicast:** One-to-many

(f) Explain the concept of Diffie-Hellman key generation. Generate public and private key pairs using RSA algorithm using 7 and 11 as two prime numbers.

Ans :

1 Diffie–Hellman Key Exchange Concept

Definition:

The **Diffie–Hellman (DH)** key exchange algorithm is a **cryptographic protocol** that allows two parties to **generate a shared secret key** over an **insecure communication channel** — without directly transmitting the key itself.

This shared key can later be used for **symmetric encryption** (like AES) to ensure secure communication.

Concept Overview:

1. Two users (say **Alice** and **Bob**) agree publicly on:
 - A **prime number (p)**
 - A **primitive root (g)** of that prime
2. Each chooses a **private key**:
 - Alice chooses private key **a**
 - Bob chooses private key **b**
3. Each computes their **public key**:
 - Alice computes ($A = g^a \mod p$)
 - Bob computes ($B = g^b \mod p$)
4. They exchange these public keys over the network.
5. Each side computes the **shared secret key**:
 - Alice computes ($K = B^a \mod p$)
 - Bob computes ($K = A^b \mod p$)

👉 Both get the **same key K**, since

[

$$B^a \bmod p = (g^b)^a \bmod p = (g^a)^b \bmod p = A^b \bmod p$$

✓ Example:

Let's take:

- ($p = 23$)
- ($g = 5$)

Alice:

- Private key ($a = 6$)
- Public key ($A = 5^6 \bmod 23 = 15625 \bmod 23 = 8$)

Bob:

- Private key ($b = 15$)
- Public key ($B = 5^{15} \bmod 23 = 19$)

Exchange:

- Alice sends 8 to Bob
- Bob sends 19 to Alice

Shared Key:

- Alice: ($K = 19^6 \bmod 23 = 2$)
- Bob: ($K = 8^{15} \bmod 23 = 2$)

✓ Common secret key = 2

② RSA Key Generation (using 7 and 11 as prime numbers)

Let's now generate RSA keys step by step.

Step 1: Choose two prime numbers

```
[
  p = 7, \quad q = 11
]
```

Step 2: Compute n

```
[
  n = p \times q = 7 \times 11 = 77
]
```

Step 3: Compute Euler's Totient ($\phi(n)$)

```
[
  \phi(n) = (p - 1)(q - 1) = 6 \times 10 = 60
]
```

Step 4: Choose Public Key (e)

Choose **e** such that:

```
[
  1 < e < 60 \quad \text{and} \quad \gcd(e, 60) = 1
]
```

Let's pick **e** = 7 (since $\gcd(7, 60) = 1$ )

Step 5: Compute Private Key (d)

d is the **multiplicative inverse** of **e mod $\phi(n)$** , i.e.,

```
[
  d \times e \equiv 1 \pmod{60}
]
```

Find **d**:

$$(7 \times d \bmod 60 = 1)$$

Try values:

- $7 \times 43 = 301 \rightarrow 301 \bmod 60 = 1$ ✓
So, **d = 43**

Step 6: Form Keys

- **Public Key (e, n) = (7, 77)**
- **Private Key (d, n) = (43, 77)**

Step 7: Example Encryption / Decryption

Let's take message (M = 5)

Encryption:

$$[\\ C = M^e \bmod n = 5^7 \bmod 77 = 78125 \bmod 77 = 47 \\]$$

Decryption:

$$[\\ M = C^d \bmod n = 47^{43} \bmod 77 = 5 \\]$$

✓ Message recovered successfully!

✓ Final Summary

Algorithm	Purpose	Key Idea
Diffie-Hellman	Securely generates a shared secret key over a public channel.	Uses modular exponentiation with a prime and primitive root.

RSA	Public-key cryptography used for encryption and digital signatures .	Uses product of two primes (p, q).
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RSA Keys (with p=7, q=11):

- Public key: (7, 77)
- Private key: (43, 77)

(g) Differentiate between PSK and FSK modulation techniques. Explain the term “Quantization”.

Ans :

1] Difference between PSK and FSK Modulation Techniques

Both **PSK (Phase Shift Keying)** and **FSK (Frequency Shift Keying)** are **digital modulation techniques** used to transmit digital data over analog channels by varying a property of a carrier wave.

(a) PSK – Phase Shift Keying

- In **PSK**, the **phase** of the carrier signal is changed according to the digital data (0s and 1s).
- The **frequency and amplitude** remain constant.
- Each unique phase represents a specific bit or group of bits.

✓ Example:

- In **Binary PSK (BPSK)**:
 - Bit ‘0’ → 0° phase
 - Bit ‘1’ → 180° phase

✓ Signal Equation:

[

$$s(t) = A \cos(2\pi f_c t + \phi)$$

]

where ϕ changes with the data bit.

(b) FSK – Frequency Shift Keying

- In **FSK**, the **frequency** of the carrier signal is varied according to the digital data.
- The **amplitude and phase** remain constant.
- Two or more frequencies are used to represent bits.

✓ Example:

- In **Binary FSK (BFSK)**:
 - Bit '0' → Frequency (f_1)
 - Bit '1' → Frequency (f_2)

✓ Signal Equation:

[

$$s(t) = A \cos(2\pi f_i t)$$

]

where (f_i) = (f_1) or (f_2), depending on the bit.

(c) Comparison Table

Feature	PSK (Phase Shift Keying)	FSK (Frequency Shift Keying)
Parameter varied	Phase of carrier signal	Frequency of carrier signal
Amplitude	Constant	Constant
Bandwidth requirement	Narrow bandwidth	Larger bandwidth
Noise immunity	More immune to noise	Less immune to noise
Power efficiency	High	Moderate

Implementation	Slightly complex	Simpler to implement
Example type	BPSK, QPSK	BFSK, MFSK

✓ In short:

PSK changes the *phase* of the carrier to represent data.

FSK changes the *frequency* of the carrier to represent data.

2 Quantization

Definition:

Quantization is the process of **converting a continuous range of amplitude values** (analog signal) into a **finite set of discrete values** during **Analog-to-Digital Conversion (ADC)**.

In simple words:

Quantization means **rounding off** continuous amplitude values to the **nearest fixed digital level**.

Steps of Conversion (in ADC):

1. **Sampling** – measure the signal at discrete time intervals.
2. **Quantization** – approximate each sample to the nearest quantized level.
3. **Encoding** – represent each quantized value in binary form.

Example:

If an analog sample value = 3.4 V,
and available quantization levels are 3 V and 4 V,
then it is **quantized to 3 V**.

Types of Quantization:

- **Uniform Quantization:** All steps have equal size.
- **Non-uniform Quantization:** Step size varies (used in speech compression).

Quantization Error:

The difference between the **actual analog value** and the **quantized value** is called **quantization error**.

It introduces a small amount of **noise** into the digital signal.

✓ In short:

Term	Explanation
Quantization	Process of mapping continuous amplitude values to discrete digital levels.
Purpose	To digitize analog signals for computer processing or transmission.
Error	Quantization noise due to rounding.

(h) Draw IPv4 header structure and explain the significance of Fragment offset.

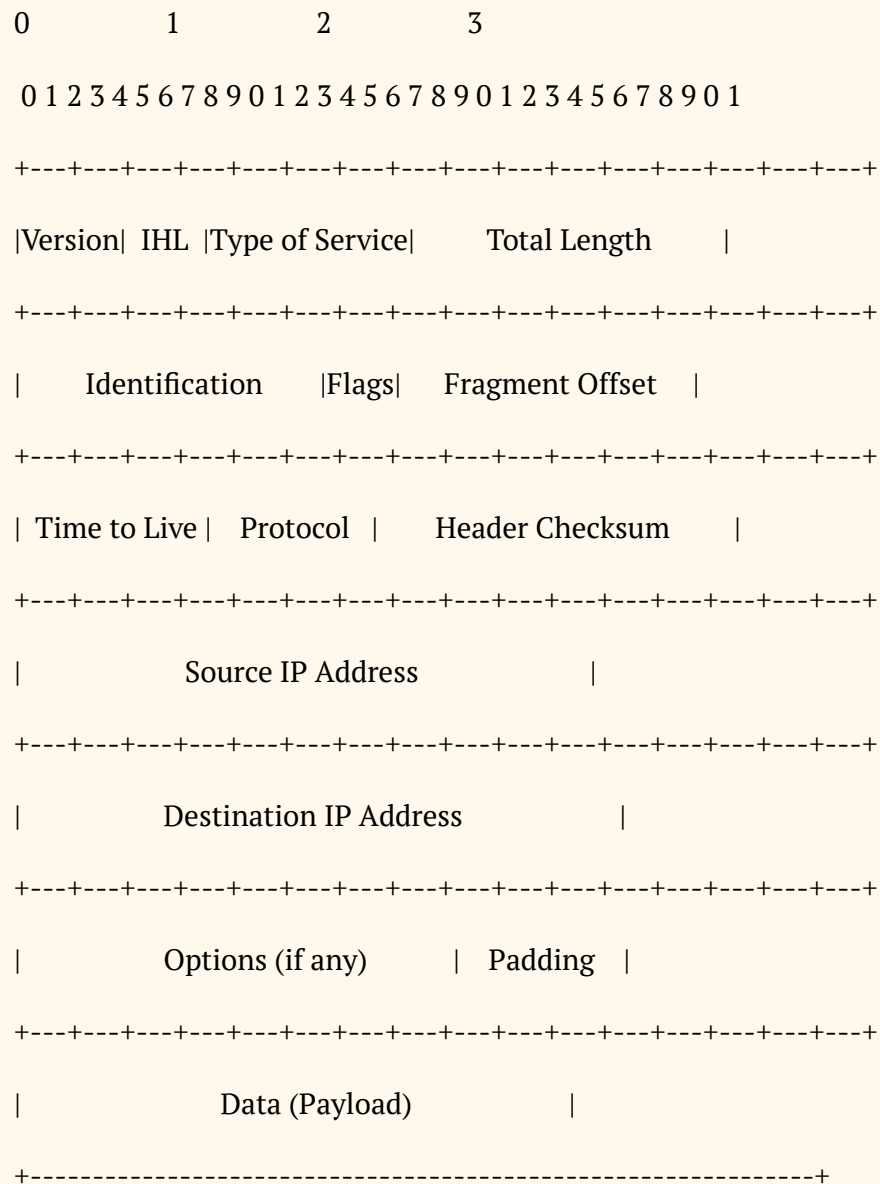
Ans :

1 IPv4 Header Structure (Diagram + Explanation)

The **IPv4 header** is a **20–60 bytes** long data structure that contains essential information used for routing and delivering packets across networks.

IPv4 Header Format

Here's a labeled representation (you can draw this in your notebook):



2 Important Fields (Brief Overview)

Field	Size	Description
-------	------	-------------

Version	4 bits	IPv4 = version 4
IHL (Header Length)	4 bits	Length of header (in 32-bit words)
Type of Service (TOS)	8 bits	Quality of service (priority)
Total Length	16 bits	Total size of the packet (header + data)
Identification	16 bits	Unique ID to identify fragments of the same packet
Flags	3 bits	Control fragmentation (DF, MF bits)
Fragment Offset	13 bits	Position of a fragment in the original packet
Time to Live (TTL)	8 bits	Limits packet lifetime
Protocol	8 bits	Indicates upper-layer protocol (e.g., TCP=6, UDP=17)
Header Checksum	16 bits	Error checking for the header
Source IP Address	32 bits	Sender's IP address
Destination IP Address	32 bits	Receiver's IP address
Options & Padding	Variable	Optional control information
Data	Variable	Actual payload being carried

3 Significance of Fragment Offset

What is Fragmentation?

When an IP packet is **too large** to travel across a network segment (whose **MTU — Maximum Transmission Unit** — is smaller), it must be **divided (fragmented)** into smaller packets.

Each fragment is transmitted separately and reassembled at the destination.

Fragment Offset Field

- The **Fragment Offset** is a **13-bit field** in the IPv4 header.
- It indicates **the position (offset)** of the fragment's data relative to the **beginning of the original packet's data**.

Key Points:

Property	Description
Size	13 bits
Unit	Offset measured in 8-byte blocks (64 bits)
Purpose	Helps the receiver reassemble fragments in the correct order
Range	0 to 8191 ($\times 8 = 65,528$ bytes max data)
Example	If offset = 100 $\rightarrow$ means the fragment starts at byte $100 \times 8 = \mathbf{800 \text{ bytes}}$ into the original data.

Example:

Suppose an original IP packet of 4000 bytes must be sent through a network with MTU = 1500 bytes.

Then:

- Fragment 1: Bytes 0–1479 → **Offset = 0**
- Fragment 2: Bytes 1480–2959 → **Offset = 185 (1480 ÷ 8)**
- Fragment 3: Bytes 2960–3999 → **Offset = 370 (2960 ÷ 8)**

At the destination, the receiver uses these offsets to **reassemble** the fragments into the original 4000-byte packet.

✓ In short:

Fragment Offset indicates the position of a fragment's data in the original packet (measured in 8-byte units).

It is essential for **reassembling fragmented IP packets** correctly at the destination.

Qus 2.

(a) What is encoding ? Explain digital-to-digital encoding with an example.

Ans :

1) What is Encoding?

Encoding is the process of **converting data from one form to another** for the purpose of **efficient transmission, storage, or processing**.

In data communication, **encoding** refers to the method of **transforming digital data (0s and 1s)** into **signals** that can be transmitted over a medium (like a wire or fiber).

💡 In simple words:

Encoding = Converting bits into signals suitable for transmission.

② Types of Encoding Techniques

In data communication, there are mainly three types of encoding:

1. **Digital-to-Digital Encoding** → Digital data → Digital signal
2. **Digital-to-Analog Encoding** → Digital data → Analog signal
3. **Analog-to-Digital Encoding** → Analog data → Digital signal

We'll focus on the **first type** 📌

③ Digital-to-Digital Encoding

Definition:

Digital-to-Digital encoding is a process of representing **digital data (binary bits)** using **discrete voltage levels** in the form of a **digital signal**.

For example,

- Logic '1' may be represented by **+5V**,
- Logic '0' may be represented by **0V** or **-5V**,
depending on the encoding scheme.

Why is it needed?

Computers generate digital data (bits), but communication channels transmit **electrical signals**.

Hence, digital data must be **encoded** into digital signals before transmission.

④ Types of Digital-to-Digital Encoding Techniques

Category	Techniques	Description
Unipolar	NRZ (Non-Return-to-Zero)	Uses only one voltage level (e.g., 0 and +V).

Polar	NRZ-Level, NRZ-Inverted, RZ	Uses both positive and negative voltages.
Bipolar	AMI (Alternate Mark Inversion), Pseudoternary	Uses three levels: +V, 0, and -V.

5 Example: NRZ-Level (Non-Return-to-Zero Level)

In NRZ-Level encoding,

- Binary '1' → Positive voltage (+V)
- Binary '0' → Zero voltage (0V)

Example:

Let's encode the bitstream: 1 0 1 1 0 0 1

Bit	Signal Level (V)
1	+5V
0	0V
1	+5V
1	+5V
0	0V
0	0V
1	+5V

Waveform (conceptually):

+5V ——— ——— ———

1	0	11	00	1

⑥ Advantages of Digital-to-Digital Encoding

- ✓ Simple and easy to implement
- ✓ Less affected by noise than analog signals
- ✓ Suitable for digital transmission systems (LANs, computers, etc.)

⑦ Disadvantages

- ✗ Synchronization required between sender and receiver
- ✗ Long sequences of 0s or 1s can cause loss of synchronization
- ✗ Some schemes require more bandwidth

✓ In short:

Term	Meaning
Encoding	Conversion of data into suitable signal form for transmission.
Digital-to-Digital Encoding	Representing digital data as discrete voltage levels.
Example (NRZ-Level)	'1' = +5V, '0' = 0V

(b) Explain the characteristics of Wide Area Network (WAN). Differentiate between client-server and peer-to-peer architecture.

Ans :

① Characteristics of Wide Area Network (WAN)

Definition:

A **Wide Area Network (WAN)** is a type of computer network that spans a **large geographical area**, connecting multiple smaller networks such as **LANs (Local Area Networks)** or **MANs (Metropolitan Area Networks)**.

It allows organizations or users in different cities, countries, or even continents to **communicate and share data**.

Key Characteristics of WAN

Characteristic	Explanation
1. Large Geographical Coverage	WAN covers large areas like cities, countries, or even the globe. Example: the Internet.
2. Interconnection of LANs and MANs	Connects multiple local or metropolitan networks together using routers, modems, or leased lines.
3. Use of Public Transmission Lines	Uses public telecommunication systems such as telephone lines, satellite links, or fiber optics.
4. Slower Speed (Compared to LAN)	Due to long-distance data transmission and multiple routing devices.
5. High Setup and Maintenance Cost	Requires expensive infrastructure, leased communication lines, and management.
6. Use of Routers and Gateways	WAN uses routers, gateways, and modems to connect distant networks.

- 7. Distributed Ownership** Usually owned by multiple organizations or service providers (e.g., ISPs).
- 8. Communication Links** Data travels through various media — optical fiber, satellite, or radio waves.
- 9. Example** The **Internet** is the largest example of a WAN.

 In short:

A WAN connects multiple LANs over large distances using routers and public communication channels like telephone networks, optical fibers, or satellites.

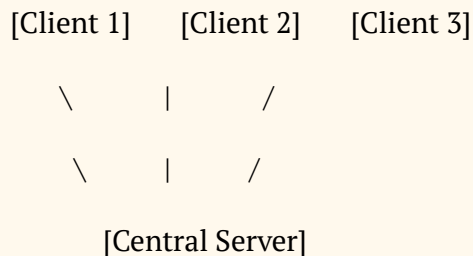
Difference Between Client–Server and Peer-to-Peer Architecture

Feature	Client–Server Architecture	Peer-to-Peer (P2P) Architecture
1. Definition	Network model where one computer acts as a server providing services, and others act as clients requesting services.	Network model where each computer (peer) can act as both client and server .
2. Control	Centralized control – the server manages all resources and security.	Decentralized control – all peers are equal and manage their own resources.
3. Data Storage	Data and files stored centrally on the server.	Each peer stores its own data locally.
4. Security	Easier to manage security through the central server.	Security is weaker because every peer manages itself.

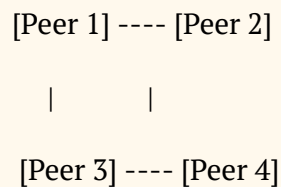
5. Performance	High performance for large networks.	Performance decreases as the number of peers increases.
6. Cost	More expensive – requires a dedicated server and maintenance.	Cost-effective – no need for a dedicated server.
7. Reliability	Server failure can affect the whole network.	If one peer fails, others can still work.
8. Examples	Email servers, Web servers, Database servers.	File sharing (BitTorrent), LAN gaming, small office/home networks.

3 Example Diagrams (for understanding)

Client–Server Model:



Peer-to-Peer Model:



✓ Summary

Aspect	WAN	Client-Server vs P2P
WAN	Large area network connecting multiple LANs using public lines (Internet).	
Client-Server	Centralized model with dedicated server managing clients.	
Peer-to-Peer	Decentralized model where each peer is both client and server.	

(c) Discuss the importance of multiplexing. List the basic multiplexing techniques.

Ans :

1] What is Multiplexing?

Multiplexing is a technique used in communication systems to **combine multiple signals** (data, voice, video, etc.) and **transmit them simultaneously** over a **single communication channel or medium**.

💡 In simple words:

Multiplexing = Sharing one communication channel among multiple signals/users.

Example:

Imagine 3 users need to send data to 3 destinations.

Without multiplexing, each user would need a separate cable.

With multiplexing, **one cable** can carry all 3 signals — saving cost and bandwidth.

2] Importance of Multiplexing

Point	Explanation
1. Efficient Use of Bandwidth	Maximizes utilization of available bandwidth by combining multiple signals.
2. Cost Effective	Reduces the need for multiple communication lines, lowering setup and maintenance costs.
3. Increases Transmission Efficiency	Allows simultaneous transmission of multiple signals through a single medium.
4. Simplifies Network Design	Reduces the number of cables and hardware needed for communication.
5. Supports Multiple Users	Enables many users to share the same channel at the same time.
6. Reduces Interference and Crosstalk	Each signal is assigned a specific channel or time, minimizing overlap.
7. Enhances System Scalability	New users can be easily added without laying new cables.

③ Basic Multiplexing Techniques

There are **three basic types** of multiplexing techniques:

Technique	Full Form	Basic Idea
1. FDM	Frequency Division Multiplexing	Different signals are transmitted at different frequency bands within the same channel.

2. TDM	Time Division Multiplexing	Each signal is assigned a specific time slot on a shared channel.
3. WDM	Wavelength Division Multiplexing	Used in optical fiber , where multiple signals are transmitted using different light wavelengths .

(a) Frequency Division Multiplexing (FDM)

- The total bandwidth is divided into **non-overlapping frequency bands**.
- Each user gets a separate frequency range.
- Used in **radio broadcasting, cable TV, and satellite systems**.

Example:

|--Ch1 (60-70 MHz)--|--Ch2 (70-80 MHz)--|--Ch3 (80-90 MHz)--|

(b) Time Division Multiplexing (TDM)

- Each user gets a **specific time slot** in a repeated sequence.
- The channel is shared **in time**, not in frequency.
- Used in **digital telephony and computer networks**.

Example:

|--User1--|--User2--|--User3--|--User1--|--User2--|--User3--|

(c) Wavelength Division Multiplexing (WDM)

- Used in **optical fiber communication**.
- Each data stream is transmitted with a **different wavelength (color of light)**.
- Allows high-speed, long-distance data transmission.

Example:

|-- λ_1 (red)--|-- λ_2 (green)--|-- λ_3 (blue)--|

4 Summary Table

Technique	Domain	Basis of Multiplexing	Used In
FDM	Analog	Frequency	Radio, TV, Cable
TDM	Digital	Time	Telephony, Computer Networks
WDM	Optical	Wavelength	Fiber Optic Communication

✓ In short:

Multiplexing increases efficiency by allowing multiple signals to share the same channel.

Basic techniques:

- FDM → Frequency-based
- TDM → Time-based
- WDM → Wavelength-based

Qus 3. (a) What is checksum ? Explain the features of sliding window protocol.

Ans :

Definition:

A **checksum** is an **error detection technique** used in data communication to detect errors in transmitted messages.

It works by generating a **small-sized numerical value (checksum value)** from the data, which helps the receiver verify whether the data was received correctly or got corrupted during transmission.

How Checksum Works:

1. Sender Side:

- The sender divides the data into **equal-sized segments (words)**.
- All segments are **added together** using **binary addition**.
- The **sum is complemented** (1's complement).
- This final value is called the **checksum**.
- The sender sends **data + checksum** to the receiver.

2. Receiver Side:

- The receiver again adds all received segments (including checksum).
- Takes the **1's complement** of the result.
- If the result is **all 0s**, the data is **error-free**.
- If not, an **error is detected**.

Example (Simple Illustration):

Suppose data is divided into 3 segments of 8 bits each:

Segment	Binary Value
1	01010101
2	01100110
3	11001100

Step 1: Add all segments (binary addition).

Step 2: Take 1's complement of the sum → **Checksum**.

Step 3: Transmit data + checksum.

Receiver repeats the process to verify accuracy.

✓ Advantages of Checksum:

- Simple and easy to implement.
- Detects common transmission errors.
- Used in many network layers (like TCP/IP).

✗ Limitations:

- Can only **detect** errors, not **correct** them.
- May **miss some complex errors** (if two errors cancel out).

② Sliding Window Protocol

Definition:

The **Sliding Window Protocol** is a method used in **data link and transport layers** for **flow control** and **error control** during transmission between sender and receiver.

It allows multiple frames to be **sent before requiring an acknowledgment (ACK)**, improving the efficiency of data transfer.

Key Concept:

Both sender and receiver maintain a **window (a range of sequence numbers)** that defines which frames can be sent or received.

As acknowledgments are received, the window **slides forward**, allowing the next set of frames to be sent.

Features of Sliding Window Protocol

Feature	Explanation
1. Flow Control	Prevents the sender from overwhelming the receiver by controlling the number of frames in transit.

2. Efficient Utilization of Bandwidth	Multiple frames can be sent before waiting for acknowledgment — reduces idle time.
3. Error Control	If a frame is lost or damaged, only that frame (or the set in the window) is retransmitted.
4. Use of Sequence Numbers	Each frame has a unique sequence number, ensuring proper order and identification.
5. Acknowledgment Mechanism	Receiver sends ACKs to confirm received frames; the window then slides forward.
6. Bidirectional Transmission	Can work in both directions simultaneously (in duplex mode).
7. Supports Pipelining	Multiple frames are in transit at the same time, increasing throughput.

Types of Sliding Window Protocols

Type	Description
1. Go-Back-N ARQ	If one frame is lost, all frames from that point onward are retransmitted.
2. Selective Repeat ARQ	Only the specific erroneous frames are retransmitted, not all.

Example (Go-Back-N):

If window size = 4,

and sender sends frames 1, 2, 3, 4 → waits for ACK.

If ACK for frame 2 is lost → sender goes back and retransmits from frame 2 onward.

Diagram (Conceptual)

Sender Window: [1][2][3][4] → sent

Receiver ACKs: ACK1, ACK2, ACK3, ACK4

Window slides → [5][6][7][8]

✓ Advantages:

- Efficient use of communication channel.
- High throughput due to pipelining.
- Provides both error and flow control.

✗ Disadvantages:

- More complex to implement.
- Requires additional memory and computation for buffering and sequence tracking.

Summary Table

Term	Meaning
Checksum	Error detection technique using binary addition and 1's complement.
Sliding Window Protocol	Flow and error control mechanism allowing multiple frames in transit simultaneously.

(b) What is pipelining ? Explain stop and wait ARQ when 'ACK' is lost, with the help of a diagram.

Ans :

Pipelining is a technique in data communication where the **sender transmits multiple frames before waiting for an acknowledgment (ACK)** from the receiver.

💡 In simple words:

Pipelining = Sending multiple frames continuously without waiting for each ACK.

This technique increases the **efficiency and throughput** of data transmission, as the sender doesn't remain idle while waiting for ACKs.

Example:

Without pipelining (Stop-and-Wait ARQ):

→ Send Frame 1 → Wait for ACK → Send Frame 2 → Wait for ACK

With pipelining (Sliding Window):

→ Send Frame 1, Frame 2, Frame 3... continuously → Receive ACKs later

Advantages of Pipelining

Feature	Explanation
1. Increases Efficiency	Channel is kept busy continuously.
2. Reduces Idle Time	Sender doesn't wait for ACK after every frame.
3. Improves Throughput	More data is sent in less time.
4. Basis for Sliding Window Protocol	Go-Back-N ARQ and Selective Repeat ARQ use pipelining.

② Stop-and-Wait ARQ Protocol

Stop-and-Wait ARQ (Automatic Repeat reQuest) is the **simplest error control protocol**.

- The **sender transmits one frame** and **waits for an acknowledgment (ACK)** before sending the next frame.

- If the **ACK is not received within a certain time (timeout period)**, the **sender retransmits** the same frame.

Working Steps:

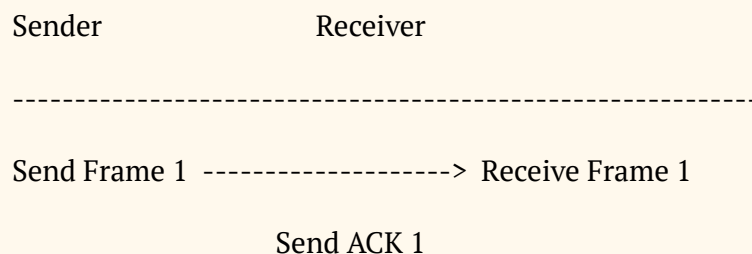
1. Sender sends **Frame 1**.
2. Receiver receives **Frame 1**, checks for errors, and sends **ACK 1**.
3. Sender waits for ACK 1 before sending **Frame 2**.
4. If ACK is **lost or delayed**, sender **resends Frame 1** after timeout.

Case: When ACK is Lost

Step-by-Step Scenario

Event	Explanation
①	Sender sends Frame 1 to Receiver.
②	Receiver receives Frame 1 correctly and sends ACK 1 back.
③	ACK 1 is lost in the transmission. Sender doesn't receive it.
④	After a timeout, sender assumes Frame 1 was lost, and retransmits Frame 1 .
⑤	Receiver gets Frame 1 again (duplicate), recognizes it, and discards it , but resends ACK 1 .
⑥	Sender finally receives ACK 1 and proceeds to send Frame 2 .

Diagram (Stop-and-Wait ARQ when ACK is Lost)



<----- (ACK 1 Lost)

[Timeout Occurs]

Resend Frame 1 -----> Receive Duplicate Frame 1

Discard Frame 1

Resend ACK 1

<----- ACK 1 Received

Send Frame 2 -----> Receive Frame 2

Explanation of Diagram:

- The **sender resends Frame 1** because it didn't get ACK 1 in time.
- The **receiver detects a duplicate** (since Frame 1 already received) and resends ACK 1.
- Communication resumes normally after the lost ACK is handled.

✓ Advantages of Stop-and-Wait ARQ

- Simple and easy to implement.
- Reliable (ensures each frame is acknowledged).

✗ Disadvantages

- Inefficient for long-distance or high-speed links (lots of waiting).
- Only one frame is transmitted at a time → **low throughput**.

Summary Table

Term

Meaning

Pipelining	Sending multiple frames without waiting for ACKs (used in Sliding Window).
Stop-and-Wait ARQ	Sends one frame and waits for ACK; retransmits if ACK is lost or timeout occurs.

(c) Briefly discuss the terms CSMA and CSMA/CD. Explain Ethernet frame format IEEE 802.3.

Ans :

1) What is CSMA?

Full Form:

CSMA = Carrier Sense Multiple Access

Definition:

CSMA is a network access method used to control how multiple devices share the same communication medium (like a cable).

 In simple terms:

Before transmitting data, each device “listens” to the channel (carrier sense) to check whether it is free.

- If the channel is **idle**, it **starts transmitting**.
- If the channel is **busy**, it **waits** until it becomes free.

Working Principle of CSMA:

1. Carrier Sense:

Each station first senses the medium to check if it's idle.

2. **Multiple Access:**

Multiple stations have access to the same channel.

3. **Collision Possibility:**

Even after sensing, two stations may transmit at the same time — causing a **collision**.

Types of CSMA:

There are four main variations based on how stations wait before retransmitting:

Type	Description
1. 1-persistent CSMA	Transmit immediately when channel is idle.
2. Non-persistent CSMA	Wait for a random time before rechecking channel.
3. p-persistent CSMA	Transmit with probability p when channel is idle.
4. CSMA/CD	Adds collision detection and retransmission mechanism.

2] What is CSMA/CD?

Full Form:

CSMA/CD = Carrier Sense Multiple Access with Collision Detection

Definition:

CSMA/CD is an enhancement of CSMA used in **wired Ethernet (IEEE 802.3)** networks.

It **detects collisions** that occur when two stations transmit simultaneously and takes **corrective action**.

Working of CSMA/CD:

1. **Carrier Sense:**
Each station listens before sending data.
2. **Transmit:**
If the medium is free, it begins transmission.
3. **Collision Detection:**
If another station transmits at the same time, both signals interfere — a **collision** occurs.
4. **Jamming Signal:**
When a collision is detected, both stations send a **jamming signal** to inform others.
5. **Backoff Algorithm:**
Each station waits for a **random time** (using exponential backoff) before trying again.

Diagram (Conceptual):

Station A: ----Transmit-----X (Collision Detected)----Wait & Retransmit

Station B: -----Transmit---X (Collision Detected)----Wait & Retransmit

Key Points:

- Used in **wired Ethernet (bus topology)**.
- Does **not work in wireless networks** (they use CSMA/CA instead).
- Reduces the chance of repeated collisions by **random backoff delays**.

③ Ethernet Frame Format (IEEE 802.3)

Definition:

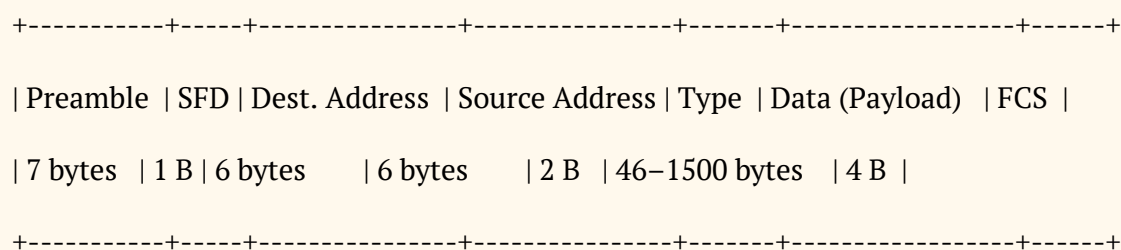
The **Ethernet frame** is the **basic data unit** used in Ethernet networks (IEEE 802.3 standard). It defines how data is **packaged, transmitted, and checked** for errors over the network.

Ethernet (IEEE 802.3) Frame Structure

Field	Size (Bytes)	Description
1. Preamble	7	Used for synchronization (alternating 1s and 0s).

2. Start Frame Delimiter (SFD)	1	Indicates the start of the frame (pattern 10101011).
3. Destination MAC Address	6	Physical (hardware) address of the receiver.
4. Source MAC Address	6	Physical address of the sender.
5. Length/Type	2	Specifies the length of data field or protocol type (e.g., IPv4, ARP).
6. Data / Payload	46 – 1500	Actual data being transmitted. If smaller than 46 bytes, padding is added.
7. Frame Check Sequence (FCS)	4	Contains CRC (Cyclic Redundancy Check) value for error detection.

Diagram of Ethernet Frame (IEEE 802.3)



Features of IEEE 802.3 Frame:

- Supports **CSMA/CD** access method.
- Ensures **error detection** using FCS.
- Uses **MAC addressing** for identifying devices.
- Minimum frame size: **64 bytes**

- Maximum frame size: **1518 bytes**

✓ Summary Table

Term	Description
CSMA	Medium access method where each device listens before transmitting.
CSMA/CD	CSMA with collision detection and retransmission (used in Ethernet).
IEEE 802.3 Frame	Standard Ethernet frame format with MAC addresses, payload, and FCS.

(d) Explain the utility of Spanning Tree and Source Routing Bridges in computer networks.

Ans :

1) What is a Bridge in Computer Networks?

A **bridge** is a **data link layer (Layer 2)** device used to **connect and filter traffic** between two or more **LAN segments**.

Its main purpose is to **reduce network congestion, divide collision domains, and forward frames** based on **MAC addresses**.

💡 Example:

A bridge connects two Ethernet LANs — only forwarding frames that need to cross between them.

2) Utility of Spanning Tree and Source Routing Bridges

There are **two main types** of bridges used in computer networks:

- **Spanning Tree Bridges** (used in Ethernet networks)

- **Source Routing Bridges** (used in Token Ring networks)

A. Spanning Tree Bridges

Definition:

A **Spanning Tree Bridge** uses the **Spanning Tree Protocol (STP)** to **prevent loops** in a network that has redundant paths.

Developed by: IEEE under standard **IEEE 802.1D**

Why Loops Are a Problem?

- In Ethernet, if multiple paths exist between bridges, **broadcast frames** can circulate indefinitely, causing a **broadcast storm**.
- Loops can lead to **duplicate frames** and **network instability**.

How Spanning Tree Works:

1. **Bridge Election:**
 - One bridge is elected as the **Root Bridge** (based on Bridge ID – lowest ID wins).
2. **Path Selection:**
 - Each non-root bridge determines the **shortest path** to the root bridge.
3. **Blocking Redundant Links:**
 - Any link that is not part of the shortest path tree is **blocked** temporarily to prevent loops.
4. **Automatic Recovery:**
 - If an active link fails, STP **recalculates** and **unblocks** an alternative path.

Diagram (Conceptual):

+-----+

| Root Bridge|

+-----+

/ \

[Bridge A] [Bridge B]

\ /

\ /

[Bridge C]

- Only loop-free paths remain active.

Key Features of Spanning Tree Bridges:

Feature	Description
Loop Prevention	Ensures a loop-free topology even with redundant paths.
Dynamic Topology Adjustment	Automatically adapts to link failures.
Redundancy	Maintains backup links that activate if a main link fails.
Standardized by	IEEE 802.1D

✓ Utility of Spanning Tree Bridges:

- Prevent **broadcast storms**.
- Enable **fault tolerance** (backup paths).
- Maintain **efficient network traffic flow**.
- Provide **automatic reconfiguration** after failures.

B. Source Routing Bridges

Definition:

Source Routing Bridges are used in **Token Ring networks (IEEE 802.5)**.

Here, the **source station (sender)** determines the **entire route** that the frame should take through the network.

💡 Unlike STP, the routing decision is made **by the sender**, not the bridge.

How Source Routing Works:

1. **Route Discovery:**
 - The sender sends a **Route Discovery frame** to find all possible paths to the destination.
2. **Response:**
 - The destination replies with a **Route Reply** frame, listing the discovered paths.
3. **Path Selection:**
 - The source node selects the **best route** (shortest, least delay, etc.) and includes it in the frame header.
4. **Frame Forwarding:**
 - Bridges forward frames **based on the route information** included by the source.

Example:

If station A wants to send to station D through bridges B1 and B2, the source routing information may look like:

A → B1 → B2 → D

This route is placed inside the frame header.

Key Features of Source Routing Bridges:

Feature	Description
Route Discovery	Sender finds and stores possible routes.

Routing Information Field (RIF)	Frame header includes route path.
Used In	Token Ring (IEEE 802.5) networks.
Bridges' Role	Forward frames according to routing information provided by source.

✓ Utility of Source Routing Bridges:

- Enables **multiple paths** between stations.
- Allows **dynamic route selection** by the sender.
- **Simplifies bridge processing** (since the bridge doesn't decide the route).
- Provides **route optimization** and redundancy.

③ Difference Between Spanning Tree and Source Routing Bridges

Feature	Spanning Tree Bridge	Source Routing Bridge
Used In	Ethernet (IEEE 802.1D)	Token Ring (IEEE 802.5)
Route Decision Made By	Bridges (automatically using STP)	Source Station (sender)
Loop Prevention	Uses tree structure to block redundant paths	Sender avoids loops by specifying route
Network Topology	Tree (loop-free)	Mesh (multiple possible routes)
Overhead	Low	Higher (due to routing info in each frame)
Failure Recovery	Automatic by STP	Manually discovered by new route discovery

✓ Summary

Bridge Type	Purpose / Utility
Spanning Tree Bridge	Prevents loops in Ethernet networks using STP and ensures redundancy.
Source Routing Bridge	Allows sender to define routes in Token Ring networks for efficient delivery.

Qus 4.

(a) What is a MAC address ? Compare virtual circuit and datagram subnets.

Ans :

1] What is a MAC Address?

Full Form:

MAC = Media Access Control

Definition:

A **MAC address** is a **unique hardware address** assigned to the **network interface card (NIC)** of a device.

It is used to **identify devices** on a **local area network (LAN)** at the **Data Link Layer (Layer 2)** of the OSI model.

💡 In simple words:

The **MAC address** is like a “**physical postal address**” for your device on the local network.

Structure of a MAC Address:

- It is **48 bits (6 bytes)** long.

- Usually written in **hexadecimal form**, separated by colons or hyphens.

Example:

00:1A:92:4F:67:89 or 00-1A-92-4F-67-89

Format Breakdown:

Part	Bits	Description
OUI (Organizationally Unique Identifier)	24 bits	Identifies the manufacturer (e.g., Intel, HP).
NIC Specific Part	24 bits	Unique number assigned to that specific device.

Features of MAC Address:

- **Unique:** No two NICs share the same MAC address.
- **Permanent:** Hard-coded into hardware by manufacturer (burned into ROM).
- **Used in LAN Communication:** Identifies devices within the same local network.
- **Layer 2 Addressing:** Works at the **Data Link Layer**, not the network (IP) layer.

Example in Use:

When your computer sends data on a LAN, it uses the **MAC address** to deliver the frame to the correct destination device (e.g., another computer or router).

2 Virtual Circuit vs Datagram Subnets

In **network layer design**, two main approaches are used to deliver packets across the network:

- 👉 **Virtual Circuit (VC) subnets**
- 👉 **Datagram subnets**

(A) Virtual Circuit Subnet

Definition:

A **virtual circuit subnet** establishes a **logical path or connection** between the sender and receiver **before** any data transfer begins.

Think of it like a **telephone call** — a connection is set up first, then data flows.

Characteristics:

Feature	Explanation
Connection-Oriented	A connection (virtual path) is established before data transfer.
Fixed Path	All packets follow the same route.
Packet Identification	Packets carry a Virtual Circuit Identifier (VCI) instead of full destination address.
Reliability	Usually more reliable; ensures packets arrive in order.
Setup Time	Initial setup time required before data transfer.
Failure Handling	If a link fails, the circuit must be re-established.
Example	Used in Frame Relay, ATM, X.25 networks .

(B) Datagram Subnet

Definition:

A **datagram subnet** sends each packet **independently**, and each packet may take a **different route** to reach the destination.

Think of it like **sending separate letters by post** — each may travel by a different route.

Characteristics:

Feature	Explanation
Connectionless	No setup phase; packets are sent independently.
Dynamic Routing	Each packet may take a different path depending on network conditions.
Packet Identification	Each packet carries the complete destination address .
Reliability	Packets may arrive out of order or get lost.
Setup Time	No initial setup time needed.
Failure Handling	Automatically adjusts to network changes (flexible).
Example	Used in IP networks (Internet) .

③ Comparison: Virtual Circuit vs Datagram Subnet

Feature	Virtual Circuit Subnet	Datagram Subnet
Connection Type	Connection-oriented	Connectionless
Route Used	Fixed path for all packets	Each packet can take a different route
Addressing	Uses Virtual Circuit Identifier (VCI)	Each packet has full destination address
Setup Phase	Required before data transfer	Not required

Reliability	Reliable; packets arrive in order	Less reliable; packets may arrive out of order
Failure Recovery	Must re-establish a new path	Automatically reroutes packets
Speed	Slower at start (setup time)	Faster to start (no setup)
Example	ATM, Frame Relay, X.25	IP, Internet

④ Diagram (Conceptual Representation)

Virtual Circuit Subnet:

Connection Setup → [A]—[B]—[C]—[D] (Path fixed)

All packets follow same route

Datagram Subnet:

[A]—[B]—[D]

[A]—[C]—[D]

[A]—[E]—[D]

Packets take different routes

✓ Summary

Term	Meaning / Use
MAC Address	Unique hardware address identifying a device on a LAN (Layer 2).

Virtual Circuit Subnet Connection-oriented network approach; all packets follow same route.

Datagram Subnet Connectionless approach; packets routed independently.

(b) Find the shortest route between points 'A' and 'E' in the graph given below :

Ans :

Find the **shortest route** between points A and E in the given weighted graph.

Given Graph Edges and Weights:

Edge Weight

A-B 1

A-D 5

B-D 2

B-E 4

D-E 1

D-C 2

C-E 3

We need: Shortest path from A → E

Let's use **Dijkstra's Algorithm** step by step.

Step 1: Initialization

Node	Distance from A	Previous Node
------	-----------------	---------------

A	0	—
---	---	---

B	∞	—
---	----------	---

C	∞	—
---	----------	---

D	∞	—
---	----------	---

E	∞	—
---	----------	---

Start from A (distance 0).

Step 2: Visit A's neighbors

From A:

- $A \rightarrow B = 1 \rightarrow \text{Distance to B} = 1$
- $A \rightarrow D = 5 \rightarrow \text{Distance to D} = 5$

Update table:

Node	Distance	Previous
------	----------	----------

A	0	—
---	---	---

B	1	A
---	----------	---

D	5	A
---	----------	---

C	∞	—
---	----------	---

E	∞	—
---	----------	---

Visited: A

Step 3: Visit B (next smallest distance = 1)

From B:

- $B \rightarrow D = 1 + 2 = 3 \rightarrow$ Better than current $D=5 \rightarrow$ update $D=3$
- $B \rightarrow E = 1 + 4 = 5 \rightarrow$ update $E=5$

Updated table:

Node	Distance	Previous
------	----------	----------

A	0	—
---	---	---

B	1	A
---	---	---

D	3	B
---	---	---

E	5	B
---	---	---

C	∞	—
---	----------	---

Visited: A, B

Step 4: Visit D (next smallest = 3)

From D:

- $D \rightarrow E = 3 + 1 = 4 \rightarrow$ better than current $E=5 \rightarrow$ update $E=4$
- $D \rightarrow C = 3 + 2 = 5 \rightarrow$ update $C=5$

Updated table:

Node	Distance	Previous
------	----------	----------

A	0	—
---	---	---

B 1 A

D 3 B

E 4 D

C 5 D

Visited: A, B, D

Step 5: Visit E (next smallest = 4)

E is destination — stop here ✓

✓ **Shortest Distance from A → E = 4**

Now let's **trace the path** using “Previous” column:

- $E \leftarrow D \leftarrow B \leftarrow A$
→ **Path: A → B → D → E**

✓ **Final Answer**

Shortest Path A → B → D → E

Total Distance 4

Verification of Alternate Routes

Route Total Distance

A-B-E $1 + 4 = 5$

A-B-D-E $1 + 2 + 1 = 4$ ✓

A-D-E $5 + 1 = 6$

A-D-C-E $5 + 2 + 3 = 10$

✓ Shortest = 4 via A-B-D-E

✓ Final Answer:

Shortest Route: $A \rightarrow B \rightarrow D \rightarrow E$

Shortest Distance: 4 units

(c) Explain Token Bucket Traffic Shaper with a suitable diagram.

Ans :

A **Token Bucket** is a **traffic shaping mechanism** used in computer networks to **control the rate of data transmission** and **allow bursts of traffic** while maintaining an average data rate.

It is commonly used in **ATM networks, Internet routers, and QoS (Quality of Service)** mechanisms.

🧠 **Concept in Simple Words:**

- Imagine a **bucket** that **collects “tokens”** at a **steady rate** (say, r tokens per second*).
- Each **token** allows the transmission of a certain number of bytes (usually 1 byte or 1 packet).
- When a packet wants to be sent:
 - If there are enough tokens in the bucket → **Packet is transmitted**.
 - If not enough tokens → **Packet waits** (or is dropped, depending on policy).
- The bucket can hold up to **b tokens (maximum capacity)**.

② Working Principle

1. Token Generation:

- Tokens are added to the bucket at a constant rate r tokens per second.

2. Bucket Capacity:

- The bucket can hold up to **b tokens** (if full, new tokens are discarded).

3. Packet Transmission:

- To send a packet of size **n**, **n tokens** are removed from the bucket.
- If **n** $\leq$ **available tokens**, packet is transmitted immediately.
- If **n** $>$ **available tokens**, the packet must **wait** until enough tokens accumulate.

4. Burst Capability:

- If the bucket has accumulated tokens (not used for a while), it allows a **burst of packets** to be sent at once — up to the limit of **b tokens**.

3 Mathematical Representation

Let:

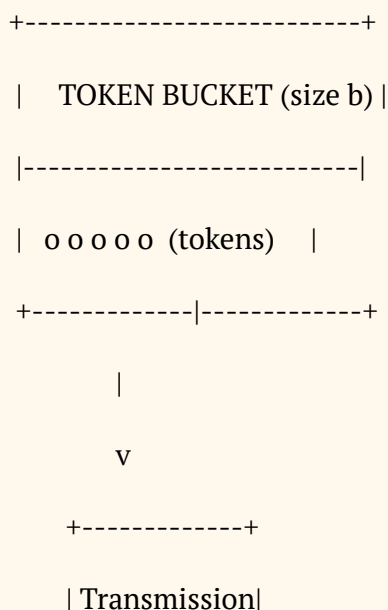
- **r** = average token generation rate (bytes/sec)
- **b** = bucket size (max tokens stored)
- **R** = maximum possible burst transmission rate

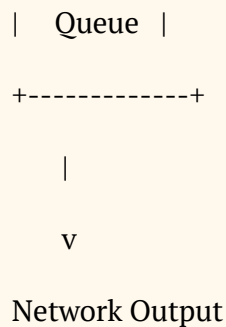
Then:

- Long-term average rate $\leq r$
- Short-term bursts allowed $\leq b$

4 Diagram of Token Bucket

Here's a conceptual diagram:





Explanation of Diagram:

- Tokens are generated at a fixed rate (r) and collected in the bucket.
- Packets arriving at the transmission queue are only sent if enough tokens are available.
- Each token removed represents permission to send one unit of data.

5 Example

Assume:

- **Token rate (r)** = 1 Mbps
- **Bucket size (b)** = 2 MB

👉 This means:

- On average, the network allows **1 Mbps**,
- But it can handle **bursts up to 2 MB** if the bucket has accumulated enough tokens.

6 Advantages of Token Bucket

Feature	Explanation
Allows Bursts	Can transmit data in bursts if tokens are available.
Flexible	Controls both average rate and peak rate.

Efficient Bandwidth Use Makes full use of available capacity.

Better QoS Provides smoother traffic flow in networks.

7 Difference Between Token Bucket and Leaky Bucket

Feature	Token Bucket	Leaky Bucket
Nature	Allows bursty traffic	Forces constant output rate
Tokens / Water	Tokens accumulate	Water leaks at steady rate
Flexibility	More flexible, allows idle-time accumulation	Less flexible, discards excess traffic
Used In	Internet traffic shaping	Traffic policing and rate control

✓ Summary

Aspect	Token Bucket Traffic Shaper
Purpose	To regulate the flow of packets in a network while allowing short bursts.
Mechanism	Uses tokens that accumulate at a constant rate and are spent when sending packets.
Control Parameters	Token generation rate (r) and bucket size (b).
Key Benefit	Allows burst transmissions while maintaining average bandwidth limit.

(d) What is meant by fragmentation ? Compare Interior and Exterior gateway routing protocols.

Ans :

Fragmentation is the process of **breaking a large IP packet into smaller packets (fragments)** so that they can be transmitted over a network that has a smaller **Maximum Transmission Unit (MTU)** size.

💡 Each network type (Ethernet, Wi-Fi, etc.) has a limit on how large a data frame can be — this limit is called **MTU (Maximum Transmission Unit)**.

When a packet is **larger than the MTU** of a network, it must be **fragmented** before transmission.

Why Fragmentation is Needed:

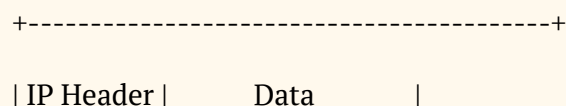
- Different networks have **different MTU values**.
- If an IP packet is too large to pass through a network link, it must be split into smaller fragments.
- This ensures that data can travel smoothly across **heterogeneous networks**.

How Fragmentation Works (IPv4):

1. The original large IP packet is divided into smaller fragments.
2. Each fragment becomes an **independent IP packet** with:
 - Its own **header**
 - **Fragment Offset** (to identify its position in the original packet)
 - **More Fragments (MF)** flag to indicate if more fragments follow
3. At the destination, the IP layer **reassembles** all fragments into the original packet.

Diagram:

Original Packet



+-----+

After Fragmentation

+-----+

| IP Header 1 | Data (part 1)|

+-----+

| IP Header 2 | Data (part 2)|

+-----+

| IP Header 3 | Data (part 3)|

+-----+

Each fragment carries:

- The **same Identification number** (to identify belonging to same packet)
- A **Fragment Offset** (position of fragment)
- The **MF bit** (set to 1 except for last fragment)

Example:

If MTU = 1000 bytes and an IP packet = 2400 bytes:

→ It will be divided into 3 fragments:

- Fragment 1 = 1000 bytes
- Fragment 2 = 1000 bytes
- Fragment 3 = 400 bytes

Key Points:

Aspect

Explanation

Done by	Routers (in IPv4) or the sending host
Reassembly done by	Destination host
Controlled by	Fields in the IP header (Identification, Fragment Offset, MF flag)
IPv6	Does not allow routers to fragment packets — done only by the source host

② Interior vs Exterior Gateway Routing Protocols

Routing protocols are used to determine **best paths** for data to travel across networks.

They are divided into two categories:

A. Interior Gateway Protocols (IGP)

Definition:

IGPs are used to exchange routing information **within a single autonomous system (AS)**.

💡 An **Autonomous System (AS)** is a group of routers under a single administrative domain (e.g., one organization or ISP).

Examples:

- **RIP (Routing Information Protocol)**
- **OSPF (Open Shortest Path First)**
- **EIGRP (Enhanced Interior Gateway Routing Protocol)**

Characteristics:

Feature	Interior Gateway Protocols (IGP)
Scope	Within one organization or AS

Routing Complexity	Simpler and faster
Metrics Used	Hop count, bandwidth, delay, etc.
Convergence Speed	High (especially OSPF)
Examples	RIP, OSPF, EIGRP
Administrative Control	Single administrative domain

B. Exterior Gateway Protocols (EGP)

Definition:

EGPs are used to exchange routing information **between different autonomous systems** (ASes).

💡 Used primarily between **ISPs** or **large networks** on the Internet.

Example:

- **BGP (Border Gateway Protocol)** — the **only standard EGP** used on the Internet today.

Characteristics:

Feature	Exterior Gateway Protocols (EGP)
Scope	Between different ASes
Routing Complexity	More complex (policy-based)
Metrics Used	Based on routing policies, not just distance
Convergence Speed	Slower (due to large scale)

Example BGP

Administrative Control Multiple organizations (different policies)

③ Comparison Table: Interior vs Exterior Gateway Protocols

Feature	Interior Gateway Protocol (IGP)	Exterior Gateway Protocol (EGP)
Area of Operation	Within a single AS	Between multiple ASes
Administrative Control	Same organization	Different organizations
Examples	RIP, OSPF, EIGRP	BGP
Routing Basis	Metric-based (hop count, cost)	Policy-based (business agreements)
Convergence Time	Faster	Slower
Routing Table Size	Smaller	Very large
Flexibility	Limited to AS	Global routing between ASes
Used For	Internal network routing	Internet-wide routing

④ Summary

Term	Meaning / Function
Fragmentation	Splitting large IP packets into smaller fragments to fit MTU limits.

IGP Routing within a single organization or AS (e.g., OSPF, RIP).

EGP Routing between different ASes on the Internet (e.g., BGP).

✓ Final Answer Summary:

- **Fragmentation** ensures large packets are split to match network MTU and reassembled at destination.
- **IGPs** manage routing **within** an organization, while **EGPs** manage routing **between** organizations or ISPs.

Qus 5.

(a) Define handshaking protocol. What are the types of services provided by the transport layer ?

Ans :

A **handshaking protocol** is a **communication process** used to **establish a connection** between two devices (sender and receiver) before actual data transmission begins.

It ensures that **both devices are ready**, agree on **communication parameters**, and can **synchronize** their data exchange.

💡 In simple words:

Handshaking is like saying “Hello, are you ready?” before starting to talk — to ensure both sides are prepared for communication.

Purpose of Handshaking:

- To **establish synchronization** between sender and receiver.
- To **agree on communication parameters** (speed, mode, sequence numbers, etc.).
- To **ensure reliable connection setup** before transmitting data.

Types of Handshaking:

(a) Two-Way Handshake

- Involves **two messages**: Request and Acknowledgment.
- Used in **simple communication systems**.
- Less reliable because it does not confirm that both sides are fully ready.

Example:

1. Sender → “Connection Request”
2. Receiver → “Acknowledgment”

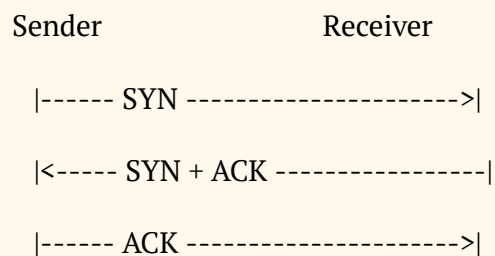
(b) Three-Way Handshake

- Used in **TCP (Transmission Control Protocol)** to establish a reliable connection.
- Ensures both sender and receiver are ready for communication.

Steps (TCP example):

1. **SYN** — Sender sends a synchronization request.
2. **SYN + ACK** — Receiver acknowledges and sends its own synchronization request.
3. **ACK** — Sender acknowledges the receiver’s request.

Diagram:



Connection Established 

After this 3-step process, both sides are synchronized and data transfer can begin.

Key Features of Handshaking Protocol:

Feature	Description
---------	-------------

Purpose	Establish and maintain connection
Reliability	Ensures both ends are ready
Used In	TCP, Bluetooth pairing, USB communication
Ensures	Synchronization, sequencing, acknowledgment

② Types of Services Provided by the Transport Layer

The **Transport Layer** (Layer 4 of the OSI Model) is responsible for **end-to-end communication** between processes running on different hosts.

It provides **logical communication** between application processes.

Main Services Provided by the Transport Layer:

(a) Connection-Oriented Service

- Establishes a connection before data transfer.
- Provides **reliable communication** (error control, flow control, retransmission).
- Example Protocol: **TCP (Transmission Control Protocol)**

Steps:

Connection establishment → Data transfer → Connection termination

Use Case: Web browsing (HTTP), Email (SMTP), File transfer (FTP)

(b) Connectionless Service

- No prior connection setup.
- Data is sent as independent packets (datagrams).
- **No guarantee** of delivery or order.
- Example Protocol: **UDP (User Datagram Protocol)**

Use Case: Streaming, online gaming, voice calls (VoIP)

(c) Reliable Data Transfer

- Ensures all data arrives at the destination **without errors** and **in correct order**.
- Uses **acknowledgments (ACKs)** and **retransmissions**.
- Handled by **TCP**.

(d) Flow Control

- Prevents sender from overwhelming receiver with too much data.
- Uses **windowing mechanism** in TCP (Sliding Window Protocol).

(e) Error Control

- Detects and corrects errors using checksums and acknowledgments.
- Retransmits lost or corrupted segments.

(f) Multiplexing and Demultiplexing

- **Multiplexing**: Allows multiple applications to use the same network connection simultaneously.
- **Demultiplexing**: Delivers received data to the correct application based on port numbers.

Example: Web (port 80), Email (port 25), DNS (port 53)

(g) Congestion Control

- Manages data transmission rate to prevent **network congestion**.
- Uses algorithms like **TCP congestion avoidance**, **slow start**, etc.

Summary Table:

Service	Description	Example Protocol
Connection-Oriented	Reliable, connection-based	TCP
Connectionless	Unreliable, no connection	UDP

Reliable Data Transfer	Ensures error-free, ordered delivery	TCP
Flow Control	Prevents receiver overload	TCP
Error Control	Detects and corrects transmission errors	TCP
Multiplexing/Demultiplexing	Supports multiple applications	TCP/UDP
Congestion Control	Prevents network overload	TCP

✅ Final Summary

Term	Explanation
Handshaking Protocol	Process of establishing synchronization and connection between sender and receiver (e.g., TCP 3-way handshake).
Transport Layer Services	Provides reliable communication, flow/error control, multiplexing, congestion control, and both connection-oriented and connectionless services.

(b) What is Nagle's Algorithm ? Explain TCP connection establishment in normal operation.

Ans :

Nagle's Algorithm is a **TCP optimization technique** used to **reduce the number of small packets** (also known as *tinygrams*) sent over the network.

It is designed to **improve network efficiency** by combining several small outgoing messages into a single, larger TCP segment.

💡 **Invented by:** John Nagle (in 1984)

Purpose:

When applications send data byte-by-byte (for example, keystrokes in telnet), this can create **many small TCP packets**, increasing:

- Overhead,
- Network congestion, and
- Inefficient bandwidth usage.

Nagle's algorithm solves this by **buffering small packets** and sending them only when:

1. The previous packet has been acknowledged, or
2. Enough data has accumulated to fill a full TCP segment.

Working Principle:

1. **First packet** is sent immediately.
2. Subsequent small packets are **held (buffered)** until:
 - The previous packet's **ACK** is received, **or**
 - The buffered data is large enough to fill a **Maximum Segment Size (MSS)**.
3. Then all buffered data is sent together in one packet.

Example:

Suppose an application sends data byte-by-byte:

A → send('H')

A → send('e')

A → send('l')

A → send('l')

A → send('o')

Without Nagle's Algorithm:

- Each `send()` results in a **small TCP segment** → lots of overhead.

With Nagle's Algorithm:

- TCP buffers the small data and sends them together as "Hello" in one packet.

Advantages:

Feature	Description
Reduces Overhead	Fewer headers, fewer ACKs
Efficient Bandwidth Use	Combines small data into larger packets
Improves Throughput	Minimizes congestion on slow links

Disadvantages:

Issue	Explanation
Increased Delay	Interactive applications (like gaming or VoIP) may experience lag
Not Ideal for Real-Time Traffic	Because it waits for ACK or enough data

⚙️ For low-latency applications, Nagle's Algorithm can be **disabled** using the TCP option `TCP_NODELAY`.

Summary of Nagle's Algorithm:

Parameter	Description
Purpose	Reduce small packet transmission
When Applied	TCP connections with small data bursts

Condition to Send Wait for ACK or full MSS

Benefit Improves efficiency, reduces congestion

Drawback Adds small delay for interactive apps

2 TCP Connection Establishment (Normal Operation)

TCP uses a **Three-Way Handshake** process to establish a reliable connection between a **client** and a **server**.

Steps of TCP Connection Establishment:

Step 1: SYN (Synchronization)

- The **client** sends a TCP segment with the **SYN (synchronize)** flag set.
- It contains the **Initial Sequence Number (ISN)** of the client.
- This tells the server that the client wants to establish a connection.

Client → Server: SYN, Seq = x

Step 2: SYN-ACK (Synchronization + Acknowledgment)

- The **server** receives the SYN, allocates buffer, and replies with:
 - A **SYN** to synchronize its own sequence number.
 - An **ACK** acknowledging the client's sequence number ($x + 1$).

Server → Client: SYN, ACK, Seq = y, Ack = $x + 1$

Step 3: ACK (Acknowledgment)

- The **client** receives SYN-ACK and responds with an **ACK** segment.
- This acknowledges the server's sequence number ($y + 1$).
- Connection is now established.

Client → Server: ACK, Seq = x + 1, Ack = y + 1

Diagram of 3-Way Handshake

Client	Server
----- SYN (Seq = x) ----->	
<----- SYN + ACK (Seq = y, Ack = x+1) -	
----- ACK (Ack = y+1) ----->	
Connection Established 	

After Connection Establishment:

- Both sides have synchronized sequence numbers.
- Data transmission begins using **TCP segments** with sequence and acknowledgment numbers to ensure **reliability**.

④ Key Features of TCP Connection Establishment

Feature	Description
Reliability	Ensures both sides are ready before communication starts
Synchronization	Both sender and receiver exchange initial sequence numbers
Bidirectional	Both sides can send and receive data after handshake
Uses Flags	SYN, ACK flags are used for control

Final Summary

Concept	Explanation
Nagle's Algorithm	Combines small TCP packets to reduce overhead and congestion.
Purpose	To improve efficiency by sending fewer, larger packets.
TCP Connection Establishment	Uses Three-Way Handshake ($\text{SYN} \rightarrow \text{SYN+ACK} \rightarrow \text{ACK}$) for reliable communication.
Result	Both sides are synchronized and ready to exchange data.

(c) What is a Feistel network ? Write short notes on Modes of Operation (CBC and OFB).

Ans :

A **Feistel network** is a **symmetric structure** used to build many block ciphers such as **DES (Data Encryption Standard)**.

It divides the input block into **two halves** and

performs multiple rounds of processing, using **subkeys** derived from the main key.

Structure of Feistel Network:

A block of plaintext (say **64 bits**) is divided into:

- **Left half (L_0)**
- **Right half (R_0)**

Each round applies a function **F** and a subkey **K_i** as follows:

Feistel Round Function:

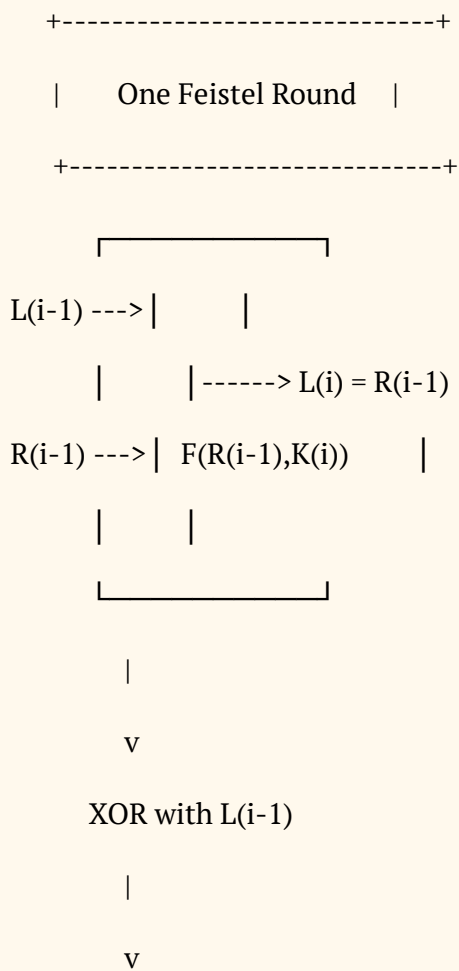
$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \text{ XOR } F(R_{i-1}, K_i)$$

Where:

- **F()** is the round function (can involve substitution, permutation, etc.)
- **K_i** is the subkey for the *i*th round
- **XOR** is the bitwise exclusive OR operation

Diagram: Feistel Structure



$$R(i) = L(i-1) \oplus F(R(i-1), K(i))$$

After **n rounds**, the two halves (L_n and R_n) are combined to form the **ciphertext**.

Decryption Process:

The same structure is used in reverse order, using subkeys in **reverse sequence** (K_n to K_1).

✓ Key Advantage:

Feistel networks use **the same algorithm** for both encryption and decryption — only the subkey order changes.

Examples of Feistel Ciphers:

- DES (Data Encryption Standard)
- Blowfish
- CAST-128
- Twofish

Features of Feistel Network:

Feature	Description
Symmetric	Same structure used for encryption and decryption
Block Size	Typically 64 or 128 bits
Rounds	Multiple rounds (e.g., 16 in DES)
Subkeys	Different key used for each round

Security Based on confusion and diffusion principles

② Modes of Operation of Block Ciphers

Block ciphers (like DES, AES) operate on **fixed-size blocks** (e.g., 64 bits or 128 bits).

To encrypt larger messages, **modes of operation** are used.

Two important modes:

👉 **CBC (Cipher Block Chaining)**

👉 **OFB (Output Feedback Mode)**

A. Cipher Block Chaining (CBC) Mode

Working Principle:

- Each **plaintext block** is **XORed** with the **previous ciphertext block** before encryption.
- An **Initialization Vector (IV)** is used for the **first block** to ensure randomness.

Encryption Process:

$$C_1 = E(P_1 \oplus IV)$$

$$C_2 = E(P_2 \oplus C_1)$$

$$C_3 = E(P_3 \oplus C_2)$$

...

Decryption Process:

$$P_1 = D(C_1) \oplus IV$$

$$P_2 = D(C_2) \oplus C_1$$

$$P_3 = D(C_3) \oplus C_2$$

Diagram: CBC Mode

$IV \xrightarrow{\oplus} E_k \rightarrow C1$
 $P1 \xrightarrow{\wedge}$
 $C1 \xrightarrow{\oplus} E_k \rightarrow C2$
 $P2 \xrightarrow{\wedge}$
 $C2 \xrightarrow{\oplus} E_k \rightarrow C3$
 $P3 \xrightarrow{\wedge}$

Features of CBC:

Feature	Description
Chaining	Each block depends on the previous ciphertext
Error Propagation	One bit error affects the current and next block
Randomness	IV ensures unique ciphertext for identical plaintexts
Use	Widely used in file encryption, TLS/SSL, etc.

B. Output Feedback Mode (OFB)

Working Principle:

- Converts a **block cipher** into a **stream cipher**.
- The **encryption function output** is fed back to generate the next keystream block.

- The **plaintext is XORed** with the keystream to produce ciphertext.

Encryption:

$$O_1 = E_K(IV)$$

$$C_1 = P_1 \oplus O_1$$

$$O_2 = E_K(O_1)$$

$$C_2 = P_2 \oplus O_2$$

...

Decryption:

(Same process as encryption)

$$P_1 = C_1 \oplus O_1$$

$$P_2 = C_2 \oplus O_2$$

Diagram: OFB Mode

$$IV \xrightarrow{E_K} O_1 \xrightarrow{\oplus} C_1$$

^ |

| P1

|

v

$$E_K \rightarrow O_2 \xrightarrow{\oplus} C_2$$

^ |

| P2

Features of OFB:

Feature	Description
Stream Cipher	Generates keystream independent of plaintext
Error Behavior	Bit errors in ciphertext affect only corresponding bits
Synchronization	Sender and receiver must stay synchronized
Use	Real-time encryption like satellite and voice communication

✓ Final Summary

Aspect	Feistel Network	CBC Mode	OFB Mode
Purpose	Cipher design structure	Mode of operation for block cipher	Mode of operation for block cipher
Input Division	Splits input into halves (L, R)	Operates on full blocks	Generates stream keystream
Dependency	Based on round function F and subkeys	Each block depends on previous ciphertext	Output depends only on previous keystream
Error Propagation	Affects entire block	2-block error propagation	Bit-level only

Common Use	DES, Blowfish	File and message encryption	Stream encryption (real-time data)
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(d) What is a Virtual Private Network (VPN) ? Write the salient features of X.509 certificates.

Ans :

A **Feistel network** is a **symmetric structure** used to build many block ciphers such as **DES (Data Encryption Standard)**.

It divides the input block into **two halves** and performs multiple rounds of processing, using **subkeys** derived from the main key.

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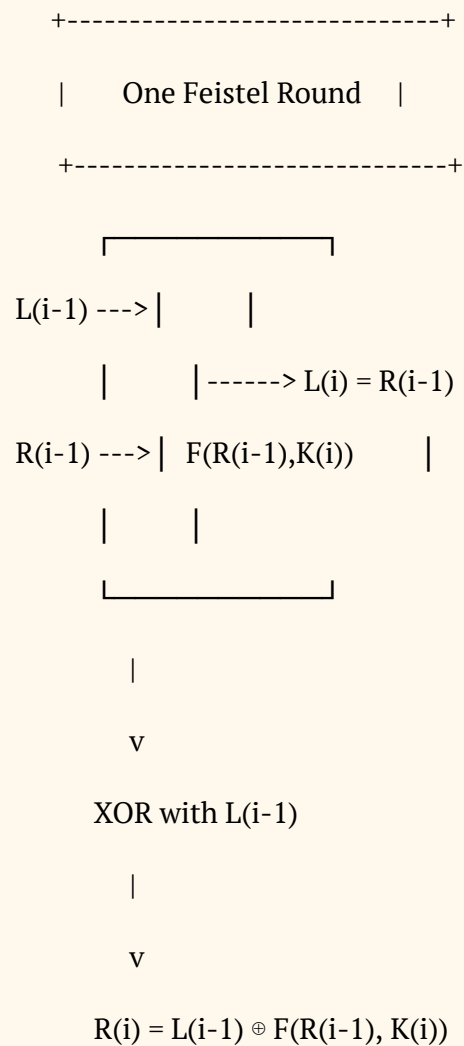
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Where:

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Diagram: Feistel Structure



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The same structure is used in reverse order, using subkeys in **reverse sequence** (K_n to K_1).

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Feistel networks use the same algorithm for both encryption and decryption — only the subkey order changes.

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Two important modes:

- 👉 CBC (Cipher Block Chaining)
- 👉 OFB (Output Feedback Mode)

A. Cipher Block Chaining (CBC) Mode

Working Principle:

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- An **Initialization Vector (IV)** is used for the **first block** to ensure randomness.

Encryption Process:

$$C_1 = E_K(P_1 \oplus IV)$$

$$C_2 = E_K(P_2 \oplus C_1)$$

$$C_3 = E_K(P_3 \oplus C_2)$$

...

Decryption Process:

$$P_1 = D_K(C_1) \oplus IV$$

$$P_2 = D_K(C_2) \oplus C_1$$

$$P_3 = D_K(C_3) \oplus C_2$$

Diagram: CBC Mode

$$IV \xrightarrow{\oplus} E_K \xrightarrow{} C_1$$

$$P_1 \xrightarrow{\wedge}$$

$C1 \xrightarrow{\oplus} E_k \rightarrow C2$

$P2 \xrightarrow{\quad} \wedge$

$C2 \xrightarrow{\oplus} E_k \rightarrow C3$

$P3 \xrightarrow{\quad} \wedge$

Features of CBC:

Feature	Description
Chaining	Each block depends on the previous ciphertext
Error Propagation	One bit error affects the current and next block
Randomness	IV ensures unique ciphertext for identical plaintexts
Use	Widely used in file encryption, TLS/SSL, etc.

B. Output Feedback Mode (OFB)

Working Principle:

- Converts a **block cipher into a stream cipher**.
- The **encryption function output** is fed back to generate the next keystream block.
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Encryption:

$$O_1 = E_K(IV)$$

$$C_1 = P_1 \oplus O_1$$

$$O_2 = E_K(O_1)$$

$$C_2 = P_2 \oplus O_2$$

...

Decryption:

(Same process as encryption)

$$P_1 = C_1 \oplus O_1$$

$$P_2 = C_2 \oplus O_2$$

Diagram: OFB Mode

IV $\xrightarrow{E_K}$ O1 $\xrightarrow{\oplus}$ C1

^ |

| P1

|

v

E_K $\xrightarrow{\quad}$ O2 $\xrightarrow{\oplus}$ C2

^ |

| P2

Features of OFB:

Feature	Description
Stream Cipher	Generates keystream independent of plaintext

Error Behavior Bit errors in ciphertext affect only corresponding bits

Synchronization Sender and receiver must stay synchronized

Use Real-time encryption like satellite and voice communication

✓ Final Summary

Aspect	Feistel Network	CBC Mode	OFB Mode
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Dependency	Based on round function F and subkeys	Each block depends on previous ciphertext	Output depends only on previous keystream
Error Propagation	Affects entire block	2-block error propagation	Bit-level only
Common Use	DES, Blowfish	File and message encryption	Stream encryption (real-time data)

Would you like me to include **a clean, exam-style labeled diagram** for the Feistel structure and both modes (CBC and OFB)? These are frequently asked in **MCS-218** 10-mark questions.

A **Feistel network** is a symmetric block-cipher structure used in ciphers such as DES.

A plaintext block is split into two halves (L_0) and (R_0).

For each round i :

```
[
 $L_i = R_{i-1}$ ,
 $R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$ 
]
```

where F is a round function and K_i a sub-key.

After all rounds, the two halves are recombined to form the ciphertext.

Using the sub-keys in reverse order decrypts the data.

Thus the same algorithm works for both encryption and decryption.

Key points

- Provides confusion + diffusion through multiple rounds.
- Easy to implement and widely used (e.g., DES, Blowfish).

Modes of Operation

Block ciphers encrypt only fixed-size blocks, so modes define how to process long messages.

1. Cipher Block Chaining (CBC)

Each plaintext block is XORed with the previous ciphertext block before encryption.

An **Initialization Vector (IV)** is used for the first block.

Encryption

```
[
 $C_1 = E_k(P_1 \oplus IV)$ ,
 $C_i = E_k(P_i \oplus C_{i-1})$ 
]
```

Decryption

```
[
 $P_1 = D_k(C_1) \oplus IV$ ,

```

```
P_i = D_k(C_i) \oplus C_{i-1}
]
```

- Chaining adds randomness and hides patterns.
- One bit error in a ciphertext block affects two plaintext blocks.

2. Output Feedback (OFB)

Turns a block cipher into a stream cipher.

A keystream is generated by repeatedly encrypting the previous output.

```
[
O_1 = E_k(IV),;
O_i = E_k(O_{i-1}),;
C_i = P_i \oplus O_i
]
```

Decryption uses the same XOR operation:

```
[
P_i = C_i \oplus O_i
]
```

- Bit errors affect only corresponding bits (no propagation).
- Sender and receiver must keep the same keystream synchronization.
- Suited to real-time or noisy links.

Summary

Feature	Feistel Network	CBC	OFB
Structure	Internal design of block cipher	Block-chaining mode	Stream-cipher mode

Dependency	Round function + subkeys	Previous ciphertext	Previous keystream
Error spread	Whole block	Two blocks	One bit
Typical use	DES, Blowfish	File/message encryption	Real-time data

These cover the concept of Feistel networks and the CBC and OFB modes of operation.

December, 2022

Qus 1.

- (a) Why bit stuffing is advantageous over character stuffing ? Write the bit sequence after bit stuffing for the data stream “1100011111110000111100”.
- (b) Differentiate between simplex, half duplex and full duplex modes of data transmission.
- (c) What is data encoding ? Explain three different ways in which encoding of analog signal with analog information is performed.
- (d) What is Pipelining ? Explain selective repeat ARQ.
- (e) Write short notes on hidden station and exposed station problem.
- (f) Explain shortest path routing algorithm with a suitable example.
- (g) What is remote procedure call ? Mention some important features of UDP.
- (h) Define a cyber threat. List some common threats in a user’s system.

Qus 2.

- (a) What are synchronous, asynchronous and isochronous communication techniques ?
- (b) What is Phase Modulation ? Why is Amplitude Modulation (AM) the most susceptible to noise ?
- (c) Define multiplexing and switching. What are the differences between ADSL and cable ?
- (d) What is Internetworking ? Differentiate between star and ring topologies of networking.

Qus 3.

- (a) Find the CRC for the data polynomial $4x^2 + x + 1$, where generator polynomial is $x^3 + 1$.
- (b) Explain stop and wait ARQ in normal operation and when frame is lost.

(c) What is slotted ALOHA protocol ? Explain its throughput calculation.

(d) Explain 802.11 protocol stack. What are source routing bridges ?

Qus 4.

(a) Compute the end to end delay for circuit switching for a network having 5 hops to switch a message of 1200 bits. Here all the links have a data rate of 4800 bps. Size of packet is 1024 bits with a header of 32 bits. Assume 0.5 sec as a call setup time and hop delay as 0.2 sec and there is no processing delay.

(b) What is distance vector routing ? Explain the count to infinity problem.

(c) Differentiate between congestion control and flow control. Explain congestion control in packet switched networks.

Qus 5.

(a) Explain the connectionless and connection oriented services provided by the transport layer.

(b) What are important features of UDP ? Why is it not considered as a reliable service ?

(c) What is a digital signature algorithm ? Explain the basis of ElGamal public key cryptosystem.

(d) What is a Firewall ? Explain the working of intrusion detection system.

June, 2023

Qus 1.

- (a) Given a signal whose amplitude varies from + 6.4 V to – 6.4 V. If we want to quantise it into 64 levels, what would be the quantised values corresponding to signals of – 3.6 V and + 0.88 V ?
- (b) What is the minimum and maximum length of the IEEE 802.3 Ethernet frame ? Differentiate between 10 Base 2 and 10 Base T ethernet cables.
- (c) List and explain policies that can be used to avoid congestion.
- (d) What is meant by public key cryptography ? Explain RSA key generation with an example.
- (e) Explain the terms Virus, Worm, Trojan and Malware.
- (f) What is noise in a signal ? Explain any three types of noise in transmission.
- (g) What is multiplexing ? Explain synchronous time division multiplexing.
- (h) Explain Bellman–Ford algorithm with a suitable example.

Qus 2.

- (a) Define transmission and propagation delays. Explain the working of fiber optic cable.
- (b) What is PCM ? Why is PAM a necessary pre-requisite to PCM ?
- (c) Differentiate between circuit switching and packet switching.
- (d) List and explain the functionality of layers in OSI reference model.

Qus 3.

- (a) Explain the terms : CRC, Error detection, Checksum, Forward error correction and Parity check.
- (b) What is Piggybacking ? Explain stop and wait ARQ with timing diagram, when ACK is lost.

(c) What is p-persistent CSMA ? Calculate the throughput of slotted ALOHA protocol.

(d) Explain the features of a transparent bridge. Discuss the operation of bridges in different LAN environments.

Qus 4.

(a) What are the important services provided by the network layer ? Compare virtual circuit and datagram approach.

(b) What is IP addressing ? Describe the address representations according to address range.

(c) Explain the features of M2M communication. Differentiate between leaky bucket and token bucket shaper.

Qus 5.

(a) List the three types of services provided by Transport layer to Application layer.

(b) How is a TCP connection established ? Explain typical three way handshake operation with a diagram.

(c) What is a Modulo Function ? Explain the principle of Elliptic curve cryptography.

(d) What is vulnerability ? List and explain Browser and Operating system related vulnerabilities.

December, 2023

Qus 1.

- (a) Define Network Topology. Compare Star and Bus topologies.**
- (b) Define Hamming Code. Write the bit stream generated by Hamming code for 001100.**
- (c) Explain the process of piggybacking with the help of an appropriate diagram.**
- (d) How does CSMA/CD differ from CSMA/CA ?**
- (e) Compare connection oriented with connectionless services.**
- (f) Define the term network congestion problem. Explain the methods to deal with it.**
- (g) Differentiate leaky bucket and token bucket traffic shaper mechanisms.**
- (h) Describe three-way handshake mechanism with the help of a diagram.**

Qus 2.

- (a) Explain ATM network technology. Write its advantages and disadvantages.**
- (b) Discuss the issues faced by the signal when it is transmitted over the transmission lines.**
- (c) Define modulation. Why is it required ? Discuss the types of modulation. Why is Amplitude Modulation (AM) most susceptible to noise ?**

Qus 3.

- (a) State the functionality of data link layer. Name different methods for framing. Give an example for each type of framing.**
- (b) What are the limitations of stop and wait flow control mechanism ? Discuss how sliding window protocol deals with their issues.**
- (c) Define vulnerable period. Draw throughput vs. load graph for pure ALOHA and slotted ALOHA. Give an expression for throughput with an assumption of no collision.**

Qus 4.

- (a) Define IP address. Compare virtual circuit with datagram subnet.
- (b) Explain Adaptive and Non-adaptive routing algorithms. Describe the concept of flooding.
- (c) Define Count to Infinity problem. How does link state routing overcome with it ? Explain the link state routing operations.
- (d) How Border Gateway Protocol (BGP) solve count to infinity problem ? Name the routers identified by OSPF.

Qus 5.

- (a) Explain Quality of Services provided by the transport layer.
- (b) Compare and contrast the flow control. Explain the flow-control and buffering mechanism at the transport-layer.
- (c) Explain Nagle's algorithm. How does it overcome the problem of wastage of bandwidth ?
- (d) Discuss Virtual Private Network (VPN) standard. Explain its types.

June, 2024

Qus 1.

- (a) Differentiate between Analog and Digital signals. Draw the diagrams for both.
- (b) Define vulnerable period. Give an expression of throughput in pure ALOHA. Also differentiate pure ALOHA with slotted ALOHA.
- (c) Define bridge. In which scenario bridge should be used ? What are the characteristics of it ?

- (d) Describe transmission and propagation delays. Explain the working of a fibre-optic cable.
- (e) List and explain the functionalities of various layers in OSI reference model.
- (f) Explain count to infinity problem with the help of an example.
- (g) Explain the features of M2M communication. Differentiate between Leaky bucket and Token bucket shaper.
- (h) Define cryptography, encryption, decryption and block-cipher. Give an example of modulo function.

Qus 2.

(a) Describe Automatic Repeat Request (ARQ). Also discuss the following methods for flow and error control :

(i) Stop and Wait ARQ

(ii) Selective Repeat ARQ

(b) Explain circuit and packet switching. Give an example where circuit switching should be applied.

Qus 3.

(a) What do you mean by error detection ? Cyclic Redundancy Check (CRC) is used to detect which type of error ? Determine CRC for the bit sequence 1101011011 where the generator polynomial key is 10011.

(b) Explain Wireless LAN protocols. What are the features of MACAW which extends MACA to improve the performance ?

(c) What is Multiplexing ? Explain synchronous time division multiplexing.

Qus 4.

(a) Explain the concept of congestion and routing in networks. Draw the graph for throughput and delay in poor and good routing.

(b) Explain the working of Dijkstra's algorithm.

(c) Define Hierarchical Routing. In which scenario, it is more advantageous ? Explain Reverse Path forwarding mechanism.

(d) Explain the mechanism on which open loop algorithm work. How is congestion controlled in packet switched network ?

Qus 5.

(a) Describe the services required by application layer from transport layer.

(b) Enlist important features of UDP. Give an example where it can be used.

(c) Explain Remote Procedure Call (RPC) with the help of an appropriate block diagram.

(d) Find the secret key using Diffie Hellman for the case, where :

User 1 : Public Key = 33, Private key = 3

User 2 : Public key = 8, Private key = 8.

And User 1 is sender, User 2 is receiver. MCS-218